

Nonintegral Tropical Degree from Two Horizontal Segments

An exact Hilbert function, explicit rational certificates,
and counterexamples to eventual polynomiality

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Abstract

For Grigoriev’s strict-witness tropical Hilbert function, with monomials of *total degree at most* k , we compute the entire function of a compact one-dimensional min-plus prevariety consisting of two horizontal segments. For every positive rational d , let

$$V_d = ([0, 1] \times \{0\}) \cup ([d + 1, d + 2] \times \{1\}).$$

We prove, for every integer $k \geq 0$,

$$\mathrm{TH}_{V_d}(k) = k + \min\{k, \lceil k/d \rceil\} + 1.$$

In particular, V_3 has tropical degree $4/3$, and its Hilbert function is a quasipolynomial of least period three, not an eventual polynomial. These give negative answers, under the definitions in arXiv:2404.06440v2, to the integrality question and the eventual-polynomiality conjecture stated in that paper’s introduction. Every rational number strictly between one and two is realized as a tropical degree by the same family. For $d \geq 1$, the formula also holds with a box degree cutoff. The proof uses only ordered slopes, a strict inequality between facing endpoints, and an explicit family of quadratic coefficient profiles. All lower bounds have rational strict witnesses. An accompanying standard-library Python verifier checks 823,974 witness inequalities in exact arithmetic, in addition to endpoint bounds and recurrence identities.

Status. The mathematical claims below have complete proofs in this draft. The proposed resolution has not been independently refereed or checked in a proof assistant. The source and novelty audit records what was actually retrieved on 20 September 2026; it is not an exhaustive priority certification.

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1 The problem and the result

1.1 The precise published target

The target is the tropical Hilbert function in Definitions 1.1 and 1.2 of Dima Grigoriev's *Degree of 1-dimensional tropical prevariety*, arXiv:2404.06440v2, revised 11 August 2025 [1]. This function counts how many bounded-degree monomials can each be the unique minimum somewhere on a fixed set, using *one common assignment of coefficients*. Its definition is reproduced carefully in Section 2.

In the introduction of that version, the author conjectures eventual polynomiality of this function and asks, as Question 2, whether the tropical degree is always an integer. The paper distinguishes total-degree and box-degree cutoffs. Our main result concerns total degree; we separately establish the relevant box-degree result in Section 7. The version matters: the original 2024 submission had the different title *A tropical version of Hilbert function* [2].

1.2 An all-degree formula

For $d \in \mathbb{Q}_{>0}$ put

$$I_0 = [0, 1] \times \{0\}, \quad I_1 = [d + 1, d + 2] \times \{1\}, \quad V_d = I_0 \cup I_1. \quad (1.1)$$

Both components have length one. Their heights differ by one, and the horizontal displacement from the right endpoint of I_0 to the left endpoint of I_1 is d .

Theorem 1.1 (Exact Hilbert function). *For every $d \in \mathbb{Q}_{>0}$ and every $k \in \mathbb{Z}_{\geq 0}$,*

$$H_d(k) := \text{TH}_{V_d}(k) = k + \min \left\{ k, \left\lceil \frac{k}{d} \right\rceil \right\} + 1. \quad (1.2)$$

In particular,

$$H_d(k) = \begin{cases} 2k + 1, & 0 < d \leq 1, \\ k + \lceil k/d \rceil + 1, & d \geq 1. \end{cases} \quad (1.3)$$

The two expressions agree when $d = 1$.

Corollary 1.2 (A counterexample with tropical degree $4/3$). *For*

$$V_3 = ([0, 1] \times \{0\}) \cup ([4, 5] \times \{1\}),$$

one has

$$\text{TH}_{V_3}(k) = k + \lceil k/3 \rceil + 1, \quad \deg_T(V_3) = \lim_{k \rightarrow \infty} \frac{\text{TH}_{V_3}(k)}{k} = \frac{4}{3}. \quad (1.4)$$

The Hilbert function is not eventually a polynomial.

The full proof of Theorem 1.1 is given in Sections 4 and 5. No asymptotic existence theorem, tropical rank theorem, or general classification of prevarieties is required for that proof. We give min-plus equations for the set itself, so membership in the source's category is explicit rather than assumed.

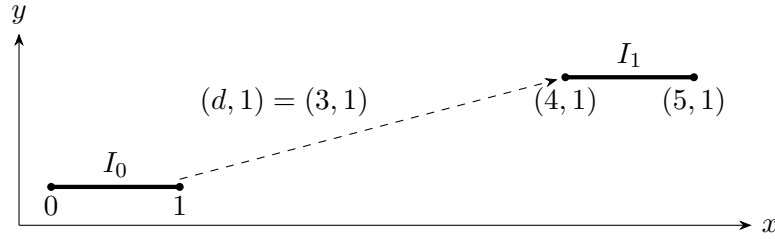


Figure 1: The counterexample V_3 . The dashed arrow is an endpoint displacement, not an additional component of the prevariety.

2 Definitions and conventions

2.1 Monomials, coefficients, and the total-degree cutoff

In min-plus notation, an ordinary affine function

$$c + ix + jy, \quad i, j \in \mathbb{Z}_{\geq 0}, \quad c \in \mathbb{R},$$

is a tropical monomial with coefficient c and total degree $i + j$. A tropical polynomial is the minimum of finitely many such affine functions. We use the finite exponent set

$$\Delta_k = \{(i, j) \in \mathbb{Z}_{\geq 0}^2 : i + j \leq k\}. \quad (2.1)$$

The coefficient-free monomial corresponding to $u = (i, j)$ is the function $x, y \mapsto ix + jy$. Additive constants are chosen when constructing a certificate; they do not create new exponent vectors.

Definition 2.1 (Strict-witness independence). A finite set $S \subseteq \Delta_k$ is independent on $V \subseteq \mathbb{R}^2$ if there exist coefficients $(c_u)_{u \in S} \in \mathbb{R}^S$ and points $(z_u)_{u \in S} \in V^S$ such that

$$c_u + \langle u, z_u \rangle < c_v + \langle v, z_u \rangle \quad \text{for every } u \in S \text{ and } v \in S \setminus \{u\}. \quad (2.2)$$

The tropical Hilbert function is

$$\text{TH}_V(k) = \max\{|S| : S \subseteq \Delta_k \text{ is independent on } V\}. \quad (2.3)$$

This is exactly the strict-inequality notion in the cited Definitions 1.1 and 1.2 [1]. We do not replace it by another notion of tropical linear independence. In particular, the coefficient vector in (2.2) is fixed while the witness point varies. Choosing a separate coefficient vector at each point would define a different and much weaker condition.

The maximum in (2.3) is over subsets of a finite set, so it is a genuine maximum. On any nonempty V , one has $\text{TH}_V(0) = 1$: the sole exponent vector is $(0, 0)$, whose singleton support is independent.

For a one-dimensional set for which the limit exists, the degree considered here is

$$\deg_T(V) = \lim_{k \rightarrow \infty} \frac{\text{TH}_V(k)}{k}. \quad (2.4)$$

Our exact formula will establish this limit directly for every V_d .

2.2 What the result does not identify

An ideal in the sense of Maclagan–Rincón has additional tropical linear-space structure [3]. Its ideal-theoretic Hilbert function is not being substituted for Definition 2.1. Likewise, a compact

union of two isolated segments is not asserted to be a balanced tropical curve. The balancedness theorem for varieties of tropical ideals belongs to a different, more restrictive setting [4].

The counterexample addresses the class actually specified in [1]: min-plus prevarieties, including finite unions of rational polyhedra. It makes no claim against an integrality theorem whose hypotheses include balancing, an ideal structure, or a different normalization.

3 Explicit equations for the prevariety

Proposition 3.1 (A three-equation presentation). *Write $d = p/q$ with positive integers p, q . The set V_d is precisely the solution set in $\mathbb{R}^2$ of*

$$\min\{2y, 1\} = y, \tag{3.1}$$

$$\min\{qx, (p+q)y\} = (p+q)y, \tag{3.2}$$

$$\min\{qx, (p+q)y + q\} = qx. \tag{3.3}$$

Every expression on either side is a tropical polynomial with nonnegative integer exponent vectors.

Proof. If $2y \leq 1$, equation (3.1) forces $2y = y$, and hence $y = 0$. If $2y \geq 1$, it forces $1 = y$, and hence $y = 1$. Conversely, both $y = 0$ and $y = 1$ satisfy it. Equations (3.2) and (3.3) are respectively equivalent to

$$qx \geq (p+q)y, \quad qx \leq (p+q)y + q.$$

Thus $y = 0$ gives $0 \leq x \leq 1$, whereas $y = 1$ gives $d+1 \leq x \leq d+2$. These are exactly the two components in (1.1). All coefficients multiplying x and y in the displayed tropical monomials are nonnegative integers; their additive constants are rational. $\square$

For the main counterexample $p = 3, q = 1$, the presentation is especially small:

$$\min\{2y, 1\} = y, \quad \min\{x, 4y\} = 4y, \quad \min\{x, 4y + 1\} = x. \tag{3.4}$$

The set is nonempty, compact, one-dimensional, and has two connected components. These are direct properties of the displayed segments.

3.1 Why a union creates a new obstruction

On a single horizontal interval, the restriction of an exponent (i, j) has slope i . Two distinct monomials with the same slope cannot both be unique minima on that interval. There are only $k+1$ possible slopes, and it is easy to realize all of them.

Two intervals give two opportunities to use the same slope with different vertical exponents. Those opportunities are not independent: the same coefficient vector must work on both components. Moving between the facing endpoints imposes a strict comparison between two monomials. The horizontal gap d then converts the finite vertical-exponent budget into a bound on the number of additional slopes that can occur. This bound contains a ceiling, and the ceiling persists at all degrees.

4 The upper bound

We first isolate the elementary facts that make the proof insensitive to degeneracies and ties.

4.1 Perturbing coefficients without losing witnesses

Lemma 4.1 (Generic endpoint winners). *Suppose S is independent on V_d . Its coefficients can be chosen so that all members of S still have strict witnesses and the minimum is attained by a unique member of S at each of the four endpoints of the two segments.*

Proof. For a fixed certificate, every difference in (2.2) is positive. When $|S| \geq 2$, let γ be the minimum of these finitely many positive differences. A perturbation of every coefficient by less than $\gamma/4$ changes any difference by less than $\gamma/2$, so it preserves every witness.

At a fixed endpoint z , a tie between distinct $u, v \in S$ is the proper affine hyperplane

$$c_u - c_v = \langle v - u, z \rangle$$

in coefficient space. The four endpoints give finitely many such hyperplanes. Their union contains no nonempty open ball. We can therefore perturb within the permitted ball while avoiding all endpoint ties. A singleton support requires no perturbation. $\square$

A monomial with a strict witness at an endpoint is also uniquely minimal on a sufficiently short one-sided interval, by continuity. Thus endpoint witnesses do not introduce exceptional zero-length pieces into the slope count below.

4.2 Counting strictly active slopes

Call a monomial *active on a segment* if it is uniquely minimal at some point of that segment. Under the generic endpoint convention, the set of active monomials includes the endpoint winners.

Lemma 4.2 (Slope order on a horizontal segment). *Along a horizontal segment traversed from left to right, the x -slopes of successive distinct active monomials strictly decrease. In particular, if the left and right endpoint slopes are u and v , the number of active monomials is at most $u - v + 1$.*

Proof. Let two distinct monomials restrict to $\alpha + it$ and $\beta + jt$. Suppose the first wins strictly at t_1 and the second at $t_2 > t_1$. Then

$$\alpha + it_1 < \beta + jt_1, \quad \beta + jt_2 < \alpha + it_2.$$

Adding gives $(i - j)(t_2 - t_1) > 0$, so $i > j$. Distinct monomials of the same slope cannot both be strictly active: their restricted difference is constant. Every active integer slope lies between the endpoint slopes, and there are $u - v + 1$ integers in that interval. $\square$

Now fix an independent support $S \subseteq \Delta_k$ and coefficients supplied by Lemma 4.1. Let S_0, S_1 be the sets active on I_0, I_1 , respectively. Because every element of S has a witness,

$$S = S_0 \cup S_1. \tag{4.1}$$

Write the four endpoint slopes in order as

	left endpoint	right endpoint
I_0	u	r
I_1	P	v .

Lemma 4.2 gives

$$|S_0| \leq u - r + 1, \quad |S_1| \leq P - v + 1, \quad 0 \leq v, r, P, u \leq k. \tag{4.2}$$

Lemma 4.3 (The universal two-segment bound). *For total degree at most k ,*

$$|S| \leq 2k + 1.$$

Proof. Each active set has size at most $k + 1$. If either has size at most k , the claim follows immediately from (4.1). Otherwise both have size $k + 1$, so each contains a monomial of x -slope k . The only exponent in Δ_k with that slope is $(k, 0)$. Consequently the two sets have a common element, and their union has size at most $2(k + 1) - 1$. $\square$

This is the one place in the upper-bound proof where the triangular degree cutoff does more than bound each exponent separately. With a box cutoff, slope k no longer identifies a unique exponent vector.

4.3 Comparing the facing endpoints

Let $a = (r, s)$ be the exponent winning at $z = (1, 0)$, and let $b = (P, Q)$ be the exponent winning at $w = (d + 1, 1)$. If $a = b$, then $P = r$ and $S_0 \cap S_1$ is nonempty. From (4.2),

$$|S| \leq (u - r + 1) + (r - v + 1) - 1 = u - v + 1 \leq k + 1. \quad (4.3)$$

This covers the shared-winner case completely.

Suppose next that $a \neq b$. Strictness at the two endpoints gives

$$c_a + \langle a, z \rangle < c_b + \langle b, z \rangle, \quad c_b + \langle b, w \rangle < c_a + \langle a, w \rangle.$$

Adding cancels the coefficients and yields

$$0 < \langle a - b, w - z \rangle = d(r - P) + s - Q. \quad (4.4)$$

Since $s \leq k - r$ and $Q \geq 0$,

$$d(P - r) < s - Q \leq k - r \leq k. \quad (4.5)$$

The integer $P - r$ is therefore strictly less than k/d , so

$$P - r \leq \lceil k/d \rceil - 1. \quad (4.6)$$

This statement is valid even if $P - r$ is negative. Using (4.2) and $u \leq k$, $v \geq 0$, we obtain

$$\begin{aligned} |S| &\leq |S_0| + |S_1| \\ &\leq u - r + 1 + P - v + 1 \\ &\leq k + (P - r) + 2 \\ &\leq k + \lceil k/d \rceil + 1. \end{aligned} \quad (4.7)$$

Proposition 4.4 (Upper half of the main theorem). *For all $k \geq 0$,*

$$\text{TH}_{V_d}(k) \leq k + \min\{k, \lceil k/d \rceil\} + 1.$$

Proof. For $k = 0$ the assertion was already noted. For $k \geq 1$, the shared-winner bound (4.3) is no larger than the claimed upper bound. In the other case combine (4.7) with Lemma 4.3. $\square$

Remark 4.5 (Why the strict ceiling is essential). The implication $n < t \Rightarrow n \leq \lceil t \rceil - 1$ is the source of the periodic term. Replacing (4.4) by a non-strict inequality would lose one unit whenever k/d is an integer. The generic perturbation does not create this strictness artificially: it preserves an already strict certificate while ensuring that the selected endpoints also have unique winners.

5 An explicit lower-bound certificate

We now construct a support of exactly the size in Proposition 4.4. The construction gives every coefficient and every witness as a rational number. It is therefore stronger than an existence argument based on a linear-programming feasibility test.

5.1 The parameters

Fix $k \geq 1$ and $d \in \mathbb{Q}_{>0}$. Define

$$m = \min\{k, \lceil k/d \rceil\} - 1, \quad \omega = \min\{1, d\}, \quad \sigma = \omega k - dm. \quad (5.1)$$

Then $0 \leq m < k$ and $\sigma > 0$. To check the last assertion, if $d \geq 1$, then $m = \lceil k/d \rceil - 1 < k/d$, and hence $\sigma = k - dm > 0$. If $d \leq 1$, then $m = k - 1$, so $\sigma = dk - d(k - 1) = d > 0$.

Choose the small positive rational numbers

$$\delta = \min\left\{\frac{1}{8}, \frac{\sigma}{8(m+1)}\right\}, \quad \varepsilon = \frac{\delta}{4k}, \quad \eta = \frac{\delta}{4(m+1)}, \quad (5.2)$$

and set

$$a = 1 - \delta, \quad b = d + 1 + \delta, \quad h = \omega k - \delta. \quad (5.3)$$

The letters a, b in this section are scalar parameters, not the endpoint exponent vectors used in the upper bound.

5.2 Two groups of monomials

Take the following coefficient-bearing monomials:

$$A_i(x, y) = \varepsilon i^2 - ai + ix + (k - i)y, \quad 0 \leq i \leq k, \quad (5.4)$$

$$B_j(x, y) = h - bj + \eta j^2 + jx, \quad 0 \leq j \leq m. \quad (5.5)$$

Thus the exponent sets are the diagonal row

$$\{(i, k - i) : 0 \leq i \leq k\}$$

and a short initial part of the bottom row

$$\{(j, 0) : 0 \leq j \leq m\}.$$

They are disjoint because $m < k$: the sole diagonal exponent with second coordinate zero is $(k, 0)$, which is not in the bottom-row group. Every exponent lies in Δ_k .

The total support size is

$$(k + 1) + (m + 1) = k + \min\{k, \lceil k/d \rceil\} + 1. \quad (5.6)$$

We must show that the minimum of all the functions in (5.4)–(5.5) is a certificate of independence.

5.3 Witnesses for the diagonal row

For $0 \leq i \leq k$, use

$$z_i = (a - 2\varepsilon i, 0). \quad (5.7)$$

Its horizontal coordinate satisfies

$$1 - \frac{3}{2}\delta \leq a - 2\varepsilon i \leq 1 - \delta.$$

Because $0 < \delta \leq 1/8$, each z_i is strictly inside I_0 . For another diagonal-row index ℓ ,

$$\begin{aligned} A_\ell(z_i) &= \varepsilon\ell^2 - a\ell + \ell(a - 2\varepsilon i) \\ &= \varepsilon(\ell - i)^2 - \varepsilon i^2. \end{aligned} \quad (5.8)$$

Consequently $A_i(z_i) = -\varepsilon i^2 \leq 0$, and A_i is the unique minimum within the A group.

It remains to keep all B terms above it. Put $x_{\min} = 1 - 3\delta/2$. For $0 \leq j \leq m$, using the nonnegativity of ηj^2 gives

$$\begin{aligned} B_j(z_i) &= h - (b - z_{i,x})j + \eta j^2 \\ &\geq h - (b - x_{\min})m \\ &= \omega k - \delta - (d + 5\delta/2)m \\ &= \sigma - \delta(1 + 5m/2). \end{aligned} \quad (5.9)$$

Here $b - z_{i,x} > 0$, so replacing j by m in the negative term is legitimate. The choice of δ implies

$$\delta(1 + 5m/2) \leq \sigma \frac{1 + 5m/2}{8(m+1)} \leq \frac{5\sigma}{16}.$$

Therefore

$$B_j(z_i) \geq \frac{11\sigma}{16} > 0 \geq A_i(z_i). \quad (5.10)$$

Every A_i has a strict witness against *all* other monomials.

5.4 Witnesses for the bottom row

For $0 \leq j \leq m$, use

$$w_j = (b - 2\eta j, 1). \quad (5.11)$$

Their horizontal coordinates satisfy

$$d + 1 + \frac{\delta}{2} < b - 2\eta j \leq d + 1 + \delta < d + 2.$$

(The weaker left inequality with $\leq$ would also suffice.) Thus the witnesses are strictly inside I_1 . For another bottom-row index ℓ ,

$$B_\ell(w_j) = h + \eta(\ell - j)^2 - \eta j^2. \quad (5.12)$$

It follows that B_j is the unique minimum within its own group, and $B_j(w_j) = h - \eta j^2 \leq h$.

For any $0 \leq i \leq k$,

$$A_i(w_j) = k + i(w_{j,x} - 1 - a) + \varepsilon i^2. \quad (5.13)$$

Since $w_{j,x} - 1 - a \geq d - 1 + 3\delta/2$, if $d \geq 1$ then $A_i(w_j) \geq k = \omega k$. If $0 < d \leq 1$, then

$$A_i(w_j) \geq k + i(d - 1) \geq k + k(d - 1) = dk = \omega k,$$

where the second inequality uses $i \leq k$ and $d - 1 \leq 0$. In both cases,

$$A_i(w_j) \geq \omega k = h + \delta > B_j(w_j). \quad (5.14)$$

Thus every B_j also has a strict witness against every competitor.

Proposition 5.1 (Lower half of the main theorem). *The tropical polynomial*

$$F_{k,d}(x, y) = \min\{A_0(x, y), \dots, A_k(x, y), B_0(x, y), \dots, B_m(x, y)\}$$

is a certificate of independence for its full support. Hence

$$\text{TH}_{V_d}(k) \geq k + \min\{k, \lceil k/d \rceil\} + 1.$$

Proof. Equations (5.8), (5.10), (5.12), and (5.14) verify all the strict inequalities in Definition 2.1. The support count is (5.6). $\square$

Proof of Theorem 1.1. For $k \geq 1$, combine Propositions 4.4 and 5.1. For $k = 0$, the unique coefficient-free monomial has a singleton certificate, so the formula also holds. $\square$

5.5 Quantitative strictness and certificate size

The construction supplies useful positive margins. Within the A group, the gap at a witness is at least ε ; within the B group, when it has more than one member, the gap is at least η . Across the groups, (5.10) gives at least $11\sigma/16$ for an A witness, and (5.14) gives at least δ for a B witness. Thus, for $k \geq 1$, a valid uniform lower bound for all competitor gaps is

$$\min\{\varepsilon, \eta, \delta, 11\sigma/16\} > 0. \quad (5.15)$$

Including η in this minimum is harmless when the B group is a singleton.

There are at most $2k + 1$ monomials, so direct certificate verification needs at most $(2k + 1)(2k)$ ordered strict comparisons. The generator uses $O(k)$ rational arithmetic operations. For $d = p/q$ in lowest terms, the displayed formulas give rational numbers with bit lengths bounded by $O(\log(k + 1) + \log(p + 1) + \log(q + 1))$ per coefficient or coordinate. Indeed, $m \leq k$, and the denominators are products of q and a bounded number of factors among 8, $4k$, and $m + 1$; the numerators grow at most polynomially in p, q, k . This is an explicit, polynomial-size certificate, not a limiting or numerical construction.

6 Degree, quasipolynomiality, and generating functions

6.1 Varying the gap and the degree

Corollary 6.1 (Degree formula). *For every positive rational d ,*

$$\deg_T(V_d) = 1 + \min\{1, 1/d\}. \quad (6.1)$$

Every rational ρ with $1 < \rho \leq 2$ is realized by a member of this family. The denominators of these degrees are unbounded, even though the number, lengths, and directions of the components are fixed.

Proof. Divide (1.3) by k and use $0 \leq \lceil k/d \rceil - k/d < 1$. For $1 < \rho < 2$, take $d = 1/(\rho - 1) > 1$. For $\rho = 2$, take $d = 1$. For example, $d = N$ gives degree $(N + 1)/N$ for every integer $N \geq 2$, and these fractions are in lowest terms. $\square$

The value 1 is approached as $d \rightarrow \infty$, but is not attained for finite d in this two-segment family. A single nondegenerate horizontal segment has Hilbert function $k + 1$, and hence degree one: the upper slope count is sharp by the same quadratic-profile construction used above. Thus the endpoint value itself is not problematic; it simply has a different witness.

6.2 Exact quasipolynomial periods

Definition 6.2. A function $f : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}$ is an eventual quasipolynomial of period $t \geq 1$ if, for each residue class modulo t , it agrees with a polynomial in k for all sufficiently large k in that class. An eventual polynomial is the special case $t = 1$.

Corollary 6.3 (Least period). *Let $d = p/q > 1$ in lowest terms, so $p > q \geq 1$. Then*

$$H_d(k) = k + \left\lceil \frac{qk}{p} \right\rceil + 1, \quad H_d(k+p) = H_d(k) + p + q. \quad (6.2)$$

It is a quasipolynomial of least eventual period p . In particular, it is not an eventual polynomial. Within the rational family V_d , eventual polynomiality holds exactly when $0 < d \leq 1$.

Proof. Write $k = pn + r$, $0 \leq r < p$. Formula (6.2) becomes

$$H_d(pn + r) = (p + q)n + r + \left\lceil \frac{qr}{p} \right\rceil + 1. \quad (6.3)$$

This proves quasipolynomiality with period p and the shift identity.

To prove minimality, set

$$E(k) = H_d(k) - \left(1 + \frac{q}{p}\right)k - 1 = \left\lceil \frac{qk}{p} \right\rceil - \frac{qk}{p}.$$

This bounded function is nonnegative and vanishes exactly when p divides k . Suppose H_d has an eventual quasipolynomial period t . On each residue class modulo t , its polynomial must be linear with slope $1 + q/p$: boundedness of $E(k)$ excludes both a higher degree and any different slope. Therefore $E(k)$ is eventually periodic with period t . Choose a sufficiently large multiple k of p . Since $E(k) = 0$, eventual periodicity gives $E(k+t) = 0$, which implies $p \mid t$. Thus the least eventual period is p . The assertion for $d \leq 1$ follows from $H_d(k) = 2k + 1$. $\square$

A shorter argument suffices merely to disprove eventual polynomiality. An eventual polynomial with linear growth must be affine. If it has integer values at all sufficiently large integers, its slope, being a difference of consecutive integer values, is an integer. Here the slope $1 + q/p$ lies strictly between one and two, so that is impossible.

6.3 The complete sequence for $d = 3$

Corollary 6.4. *Writing $k = 3n + r$, the main example has*

$$\begin{aligned} H_3(3n) &= 4n + 1, \\ H_3(3n + 1) &= 4n + 3, \\ H_3(3n + 2) &= 4n + 4. \end{aligned}$$

Its first values are

$$1, 3, 4, 5, 7, 8, 9, 11, 12, 13, 15, 16, 17, 19, 20, 21, \dots$$

The first differences repeat 2, 1, 1 from the start.

Proof. Substitute the three residue classes into (1.4). $\square$

The ordinary generating function has a short reduced rational form:

$$G_3(z) = \sum_{k \geq 0} H_3(k)z^k = \frac{(1+z)^2}{(1-z)(1-z^3)}. \quad (6.4)$$

For completeness, put $C(k) = \lceil k/3 \rceil$. Then $C(0) = 0$ and $C(k) - C(k-1)$ equals one precisely when $k \equiv 1 \pmod{3}$. Consequently

$$(1-z) \sum_{k \geq 0} C(k)z^k = \frac{z}{1-z^3},$$

and hence

$$G_3(z) = \frac{1}{(1-z)^2} + \frac{z}{(1-z)(1-z^3)} = \frac{1+2z+z^2}{(1-z)(1-z^3)}.$$

The numerator is nonzero at every cube root of unity, including one, so the displayed fraction is reduced. Multiplying by the denominator gives the constant-coefficient recurrence

$$H_3(k) = H_3(k-1) + H_3(k-3) - H_3(k-4), \quad k \geq 4, \quad (6.5)$$

with initial values 1, 3, 4, 5. Rationality of a generating function is thus fully compatible with failure of eventual polynomiality of its coefficients.

6.4 A rational generating function for every rational gap

For $d = p/q > 1$ in lowest terms, define the integers

$$e_r = \left\lceil \frac{qr}{p} \right\rceil - \left\lceil \frac{q(r-1)}{p} \right\rceil, \quad 1 \leq r \leq p.$$

They belong to $\{0, 1\}$ and sum to q . The increments of $\lceil qk/p \rceil$ repeat this block. Applying the same first-difference argument gives

$$\sum_{k \geq 0} H_d(k) z^k = \frac{1 + z + \cdots + z^{p-1} + \sum_{r=1}^p e_r z^r}{(1-z)(1-z^p)}. \quad (6.6)$$

This formula is not asserted to be reduced for every p, q . Its role is to give a direct rational representation and another way to generate the sequence exactly. When $0 < d \leq 1$, the generating function is simply $(1+z)/(1-z)^2$.

7 The separate box-degree convention

The source also considers

$$\square_k = \{(i, j) \in \mathbb{Z}_{\geq 0}^2 : 0 \leq i \leq k, 0 \leq j \leq k\}$$

in place of Δ_k [1]. Let $\text{TH}_V^\square(k)$ denote the resulting maximum number of independently witnessed monomials. It is important to check this modification rather than silently identifying it with total degree.

Proposition 7.1 (The same counterexample for a box cutoff). *For every rational $d \geq 1$ and $k \geq 0$,*

$$\text{TH}_{V_d}^\square(k) = \text{TH}_{V_d}(k) = k + \lceil k/d \rceil + 1. \quad (7.1)$$

Thus the box-normalized degree of V_3 is also $4/3$, and its box Hilbert function is not eventually polynomial.

Proof. The inclusion $\Delta_k \subseteq \square_k$ and the existing certificate give the lower bound. For the upper bound, apply the perturbation and slope lemmas to an independent subset of $\square_k$. The shared-winner case still gives $|S| \leq k + 1$.

For distinct facing winners, (4.4) still holds. Although we no longer have $s \leq k - r$, we still have $s \leq k$ and $Q \geq 0$. Therefore $d(P - r) < k$, which is all that is needed for (4.6) and (4.7). It follows that $|S| \leq k + \lceil k/d \rceil + 1$. For $d \geq 1$, this agrees with the total-degree lower bound. The case $k = 0$ again consists of the unique constant monomial. $\square$

No box formula for $d < 1$ is needed for either counterexample, and none is being inferred from the total-degree formula in that range. In particular, Lemma 4.3 itself is not a valid box-degree argument.

8 A seven-monomial certificate that can be checked by hand

The first few values already expose the nonconstant increments, but a small explicit certificate makes the lower-bound mechanism tangible. Take $d = 3$ and $k = 4$. The theorem predicts $H_3(4) = 7$.

Use five diagonal-row monomials

$$A_i(x, y) = \frac{i^2}{100} - \frac{9i}{10} + ix + (4 - i)y, \quad 0 \leq i \leq 4,$$

together with

$$B_0(x, y) = \frac{39}{10}, \quad B_1(x, y) = -\frac{19}{100} + x.$$

These parameters are chosen for a compact table; they are not the particular small-parameter choice in (5.2). Their strict witnesses are listed below.

Term	Exponent	Coefficient	Witness	Own value
A_0	(0, 4)	0	(9/10, 0)	0
A_1	(1, 3)	-89/100	(22/25, 0)	-1/100
A_2	(2, 2)	-44/25	(43/50, 0)	-4/100
A_3	(3, 1)	-261/100	(21/25, 0)	-9/100
A_4	(4, 0)	-86/25	(41/50, 0)	-16/100
B_0	(0, 0)	39/10	(41/10, 1)	390/100
B_1	(1, 0)	-19/100	(102/25, 1)	389/100

Table 1: An exact seven-term certificate at $k = 4$. Every witness belongs to V_3 , and every exponent has total degree at most four.

Let T_u denote a row's selected term and z_u its witness. To avoid relying on a plotted lower envelope or rounded values, Table 2 prints *all* competitor gaps, multiplied by 100:

$$G_{u,v} = 100(T_v(z_u) - T_u(z_u)).$$

The diagonal is zero and every off-diagonal entry is a positive integer. This proves all 42 strict comparisons directly, with least gap 1/100.

Witness for	A_0	A_1	A_2	A_3	A_4	B_0	B_1
A_0	0	1	4	9	16	390	71
A_1	1	0	1	4	9	391	70
A_2	4	1	0	1	4	394	71
A_3	9	4	1	0	1	399	74
A_4	16	9	4	1	0	406	79
B_0	10	231	454	679	906	0	1
B_1	11	230	451	674	899	1	0

Table 2: The complete scaled gap matrix for Table 1. Columns identify competitors, not alternative witnesses.

The endpoint upper bound gives $H_3(4) \leq 4 + \lceil 4/3 \rceil + 1 = 7$. The small example is therefore optimal, not merely a nonmaximal independent set. Its JSON representation is included in the accompanying data.

9 Exact computation and reproducibility

9.1 What was actually checked

The Python program `code/verify.py` uses only the standard library, and all numerical values relevant to certification are instances of `fractions.Fraction`. No floating-point tolerances, numerical linear programming, or external computer algebra system enter the verifier.

The recorded run used every integer $0 \leq k \leq 40$ and the fourteen values

$$\frac{1}{4}, \frac{1}{2}, \frac{2}{3}, 1, \frac{5}{4}, \frac{3}{2}, \frac{5}{3}, 2, \frac{7}{3}, 3, 4, 5, 10, \frac{101}{7}.$$

For each case it constructed the exact certificate and checked that all exponents were distinct, their degrees satisfied the cutoff, all witnesses lay in the correct segments, and every selected term was strictly smaller than every other selected term at its witness.

Recorded check	Count or result
Rational gap parameters	14
Parameter–degree cases	574
Strict witness inequalities, including the hand example	823,974
Total-degree endpoint-relaxation checks	574
Box endpoint-relaxation checks for $d \geq 1$	451
Generating-function coefficients checked for $d = 3$	121
Least competitor gap in the hand example	1/100
Overall result	PASS

Table 3: Recorded results. The finite checks supplement the all-degree proof; they do not prove it by extrapolation.

9.2 The independent endpoint-state calculation

The verifier also computes an upper-bound relaxation by enumerating integer pairs (r, P) with $0 \leq r, P \leq k$. In the total-degree calculation, it retains the necessary condition

$$d(P - r) < k - r,$$

maximizes the relaxed count $k + P - r + 2$ over these states, includes $k + 1$ for a shared endpoint winner, and caps the result by $2k + 1$. For the box calculation, it uses $d(P - r) < k$ and the elementary cap $2k + 2$. In the tested parameter range these independently computed upper bounds match the certificate cardinalities exactly.

This is *not* an exhaustive search over all coefficient vectors, all supports, or all tropical polynomials. Some enumerated endpoint states need not be realizable. Because they are only used to relax necessary conditions, that does not invalidate the upper bound. Its all-degree validity is supplied by Section 4, not by the enumeration.

9.3 Commands and files

From the unpacked package directory, run

```
python3 code/verify.py --max-k 40
python3 code/select_area.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The second command audits the saved random-selection record without drawing again. A deliberately requested new draw writes a separate output file. The verifier should not be run with Python's `-O` flag, which disables assertions; the entry point explicitly rejects that mode.

`data/verification.csv` contains one row per parameter-degree case, including the least exact gap. The JSON summary records aggregate counts. Separate JSON files contain the hand certificate and a degree-12 certificate for $d = 3$. The first 121 values of H_3 appear in `data/hilbert_d3.csv`. The source and proof audits are plain-text files under `notes/`.

10 Interpretation and limits

10.1 What has been established

The counterexample is not based on an irrational slope, irrational exponent, nonclosed set, limiting support, or ambiguous degree convention. It uses two closed rational horizontal segments, nonnegative integer exponents, ordinary total degree, and a finite family of rational coefficients at every k . The coefficients are allowed to depend on k , as they must be when evaluating a Hilbert function degree by degree. Within each fixed k , however, the same coefficients serve every witness.

The exact formula supplies both a lower and an upper bound for every degree. It therefore proves the degree limit as well as its nonintegrality. A single finite certificate would not suffice to establish either asymptotic claim; that is why the universal upper bound and uniform construction are both included.

The geometric effect is visible even though the component directions never change. Increasing the horizontal gap restricts how much the active x -slope can reset between the components. The available reset is asymptotic to k/d , and its integer rounding yields a genuine quasipolynomial. Consequently, for this invariant, the relative placement of components contains information not captured by their separate directions and lengths.

10.2 What is left outside the claim

The proof does not determine the Hilbert function of every rational polyhedral set. It does not establish a general eventual-quasipolynomiality theorem, a general rationality theorem for tropical degrees, or the existence of higher-dimensional normalized limits. Nor does it settle the same questions under a connectedness or balancing hypothesis.

The algebraic inequalities in the construction remain meaningful for a real parameter $d > 0$. We have deliberately stated the main geometric claim for rational d , so it lies within the source's rational-polyhedral setting without requiring a convention about irrational defining data. No assertion about irrational tropical degree in that rational category is being made.

10.3 A focused next mathematical question

A natural continuation is whether every finite union of rational segments has an eventually quasipolynomial strict-witness Hilbert function. The example shows that polynomiality is too strong, but does not decide this replacement. The endpoint argument suggests finite systems of integer inequalities whose right-hand sides grow with k ; proving that such endpoint data completely characterize realizable supports would be the missing step, not a consequence of the computations reported here.

A Problem selection, exclusions, and source audit

A.1 The random draw

Before selecting a tropical problem, a fixed list of 96 mathematical areas was constructed, and one call to Python’s `secrets.randbelow(96)` was made. The result was index 49 in zero-based indexing, or **50 in one-based indexing: tropical geometry**. No redraw was made. The list and the result are preserved in `data/area_selection.json`; the full list is printed in Appendix C.

An operating-system random draw is not a deterministic sequence that another reader should expect to reproduce. What is reproducible is the audit of the saved draw, its selected list entry, and the subsequent calculations. The packaged selection script defaults to this nonrandom audit; a new random draw requires an explicit command and does not overwrite the retained record.

A.2 The manifest exclusion check

The supplied `manifest(1).tex` was read in full. It catalogues 71 research packages, and none of its listed problems concerns strict-witness tropical Hilbert functions, tropical-degree integrality, or the two-segment family studied here. Its own description distinguishes the claims in its reports from independently checked proofs; for this task all listed problems were excluded regardless of that distinction.

The SHA-256 digest of the supplied manifest is

```
729fb6e3aef3f30c7c5284c4296681fba72
b6d6af68211f178ad0bc3c5d15b41
```

The line break is typographical only: the digest has 64 hexadecimal digits. The saved JSON record contains the uninterrupted digest.

A.3 Version and priority qualification

The arXiv abstract page retrieved on 20 September 2026 listed v2, dated 11 August 2025, as the latest version of 2404.06440. The v2 HTML and parsed PDF text were consulted for the definitions and questions. PDF screenshot requests failed with a fetch/cache error, so no claim is made that the source’s rendered PDF pages were visually inspected. The present article’s own compiled PDF was rendered and inspected separately.

Targeted searches for the paper title and for combinations of tropical Hilbert function, counterexample, noninteger degree, and segments did not locate a later resolution. Some searches were noisy or uninformative. This supports the limited statement that the latest primary version retrieved still poses the questions, not a guarantee that no unindexed preprint, private proof, or independent concurrent work exists.

A.4 Dependency audit

The definitions and problem statements are the only mathematical inputs needed from [1]. The degree computation does not invoke that paper’s general existence proof or its formulas for stars or individual polyhedra. References [3, 4] serve only to distinguish the ideal-theoretic setting from the question answered here. Every bound and certificate used in the result is proved within this article.

B Checklist for auditing the proof

The following are the points at which an off-by-one error or a change of convention would affect the conclusion.

1. The exponent region is $i + j \leq k$, not $i + j < k$ and not a square. The separate box result has its own proof.
2. The support is one finite set with one common coefficient vector. Every chosen monomial has a strict witness against every other chosen monomial, not merely against monomials on its own segment.
3. The perturbation is inside an open set preserving all original strict witnesses. Endpoint uniqueness is then obtained by avoiding finitely many proper hyperplanes.
4. The count on a horizontal segment uses distinct integer slopes. Terms that tie at an isolated point but never win strictly do not contribute an additional active slope.
5. The two active sets may overlap. Their union is counted, and the shared facing-winner case is separated before applying the strict comparison.
6. At distinct facing winners, adding the two strict inequalities gives $d(P - r) < s - Q$, with this sign. The strict inequality implies $P - r \leq \lceil k/d \rceil - 1$ even when k/d is integral.
7. The universal bound $2k + 1$ uses the unique total-degree exponent $(k, 0)$ of x -slope k . It must not be copied to a box cutoff.
8. In the lower bound, $m < k$ makes the two rows of exponent vectors disjoint. The slack σ is positive for both $d \leq 1$ and $d \geq 1$.
9. The witnesses lie in the interiors of the required segments. The cross-group gaps are explicitly positive, including at the transition $d = 1$ and when k/d is an integer.
10. The case $k = 0$ is handled separately before dividing by k in the certificate parameters. Limits and nonpolynomiality are then deduced from the all-degree formula, not from the finite verification range.

C The complete random-selection list

The ordering below is the ordering supplied to the random draw. Both columns use the original one-based indices; they do not define separate draws.

No.	Area	No.	Area
1	Additive combinatorics	49	Transcendental number theory
2	Algebraic combinatorics	50	Tropical geometry
3	Analytic number theory	51	Word combinatorics
4	Arithmetic geometry	52	Zero-sum theory
5	Automata and formal languages	53	Association schemes
6	Banach space geometry	54	Coding theory
7	Combinatorial commutative algebra	55	Discrete optimization
8	Combinatorial design theory	56	Extremal set theory
9	Combinatorial game theory	57	Frame theory
10	Computable analysis	58	Graph colorings
11	Computational geometry	59	Incidence geometry
12	Convex geometry	60	Infinite-dimensional holomorphy
13	Discrete geometry	61	Matrix inequalities
14	Dynamical systems	62	Moment problems

No.	Area	No.	Area
15	Enumerative geometry	63	Orthogonal polynomials
16	Ergodic theory	64	Positive definite functions
17	Extremal graph theory	65	Rational approximation
18	Finite fields	66	Riemannian geometry
19	Finite group theory	67	Simplicial complexes
20	Functional equations	68	Tensor rank
21	General topology	69	Topological dynamics
22	Geometric group theory	70	Universal algebra
23	Graph polynomials	71	Ultrametric analysis
24	Hypergraph theory	72	Valuation theory
25	Integer partitions	73	Boolean function theory
26	Integral inequalities	74	Chip-firing and sandpiles
27	Knot invariants	75	Discrete probability
28	Lattice theory	76	Fractional calculus
29	Linear preserver problems	77	Group actions on trees
30	Matroid theory	78	Hyperplane arrangements
31	Metric geometry	79	Information inequalities
32	Noncommutative algebra	80	Modular forms
33	Numerical semigroups	81	Monomial ideals
34	Operator inequalities	82	Optimal transport
35	Ordered algebraic structures	83	Partial orders
36	Permutation patterns	84	Real-rooted polynomials
37	Polynomial dynamics	85	Rewriting systems
38	Probabilistic combinatorics	86	Schubert calculus
39	Quasigroups and loops	87	Set-theoretic topology
40	Ramsey theory	88	Special functions
41	Real algebraic geometry	89	String-rewriting and monoids
42	Recurrence sequences	90	Sum-product phenomena
43	Riordan arrays	91	Topological semigroups
44	Semigroup theory	92	Visibility in graphs
45	Spectral graph theory	93	Weighted lattice paths
46	Symbolic dynamics	94	Well-quasi-orders
47	Tiling theory	95	Zeta functions of graphs
48	Topological graph theory	96	Numerical linear algebra

References

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